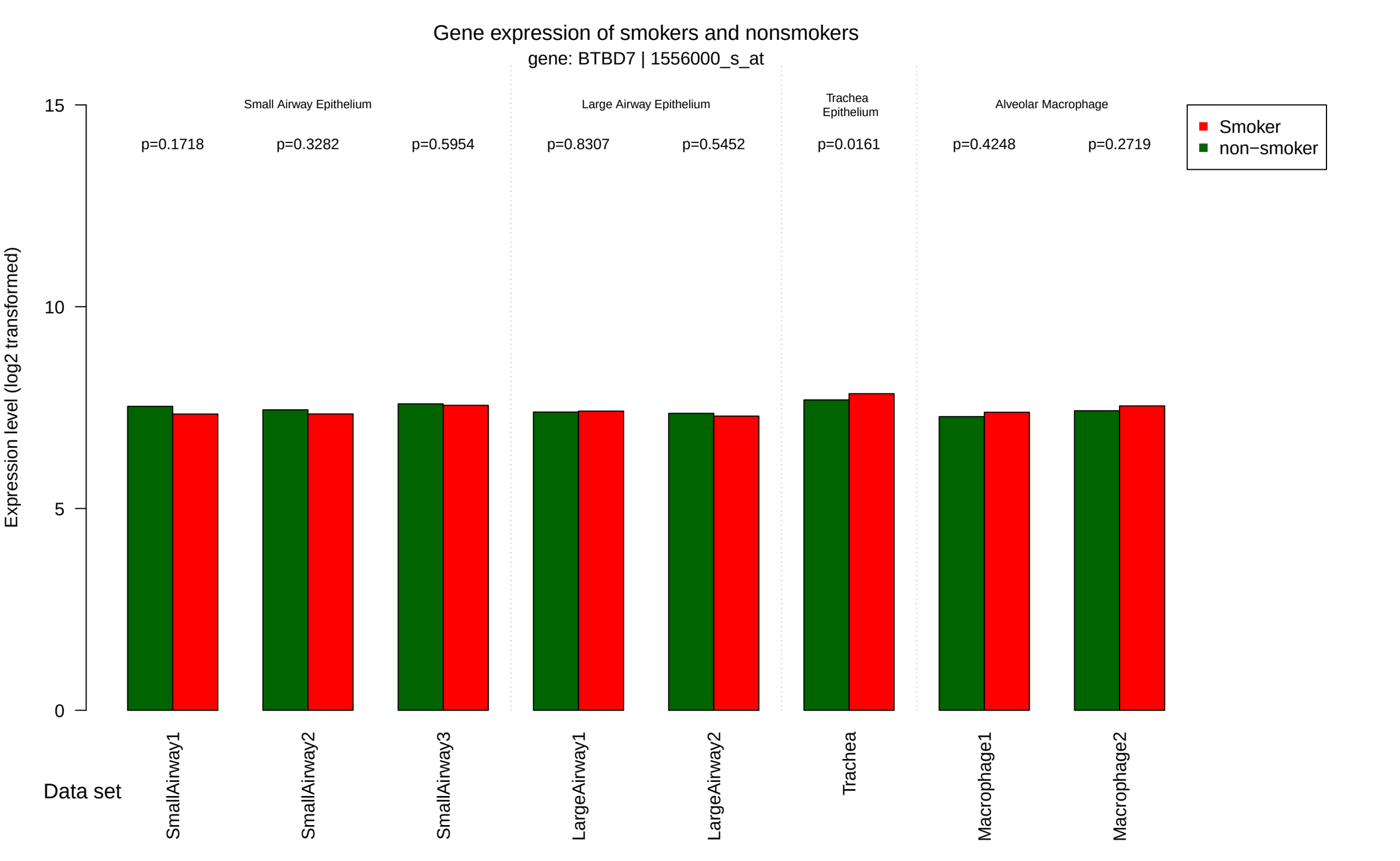
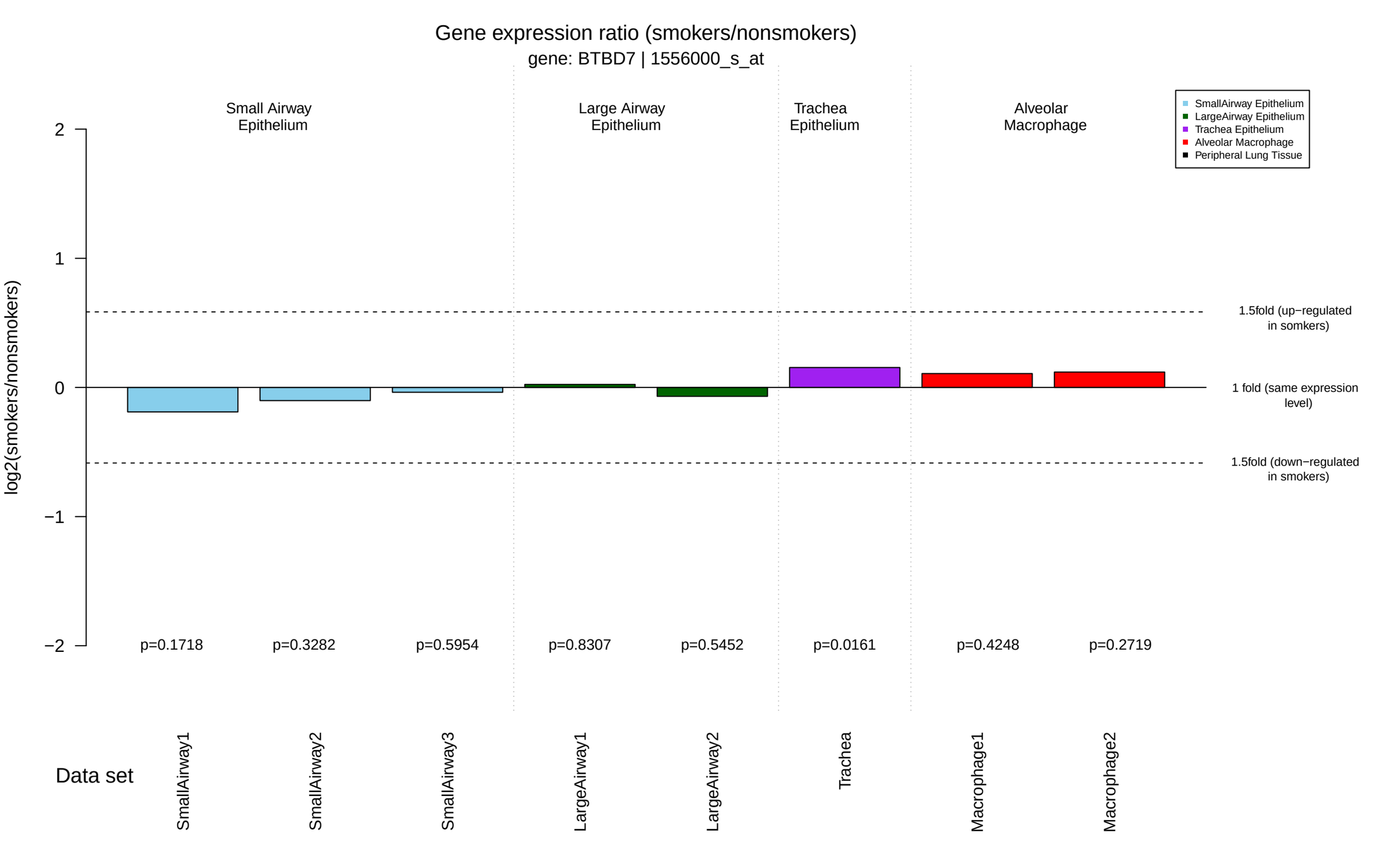
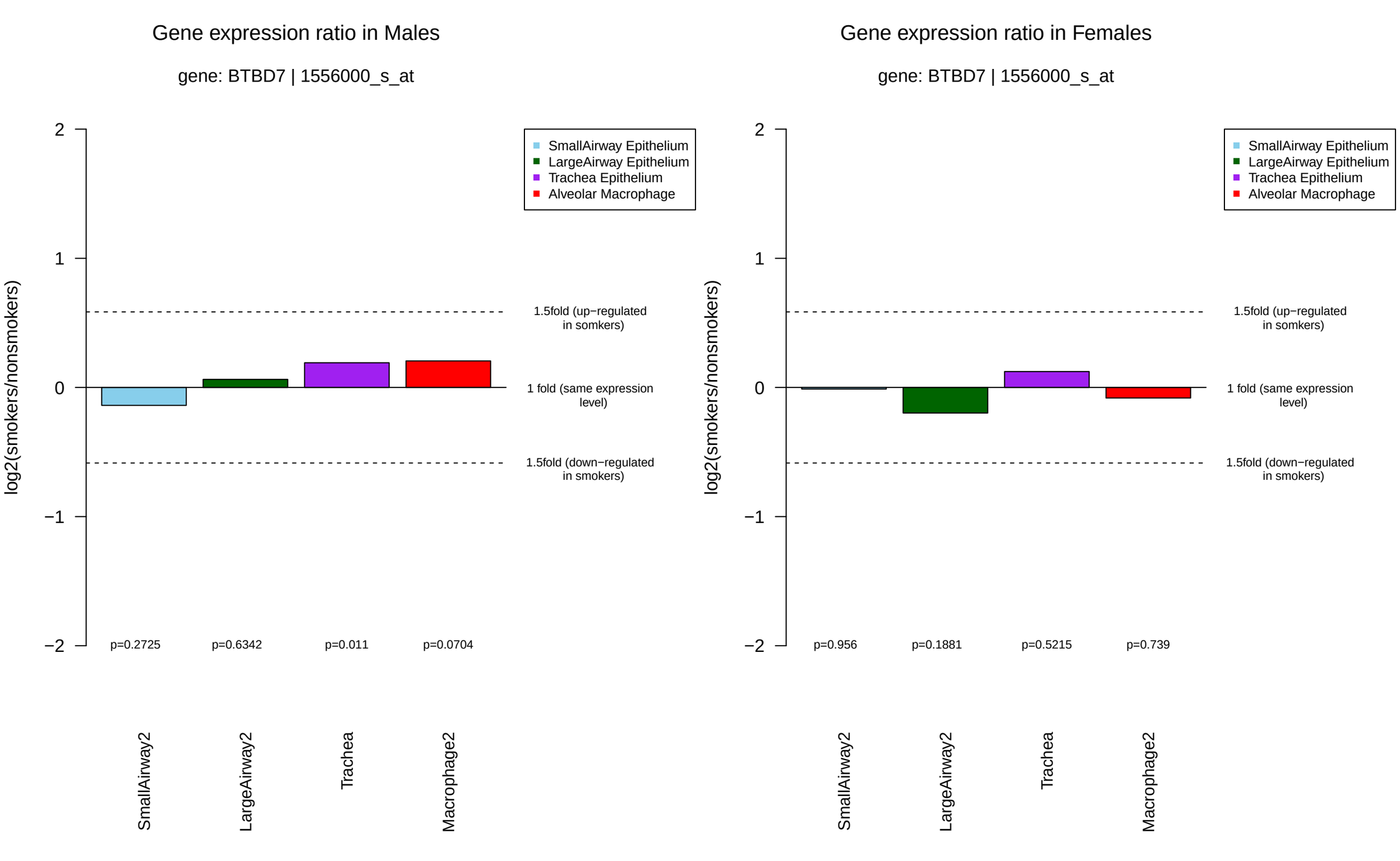
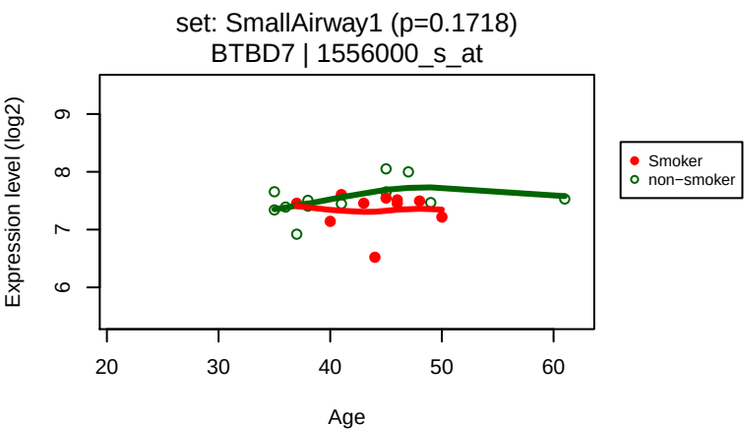
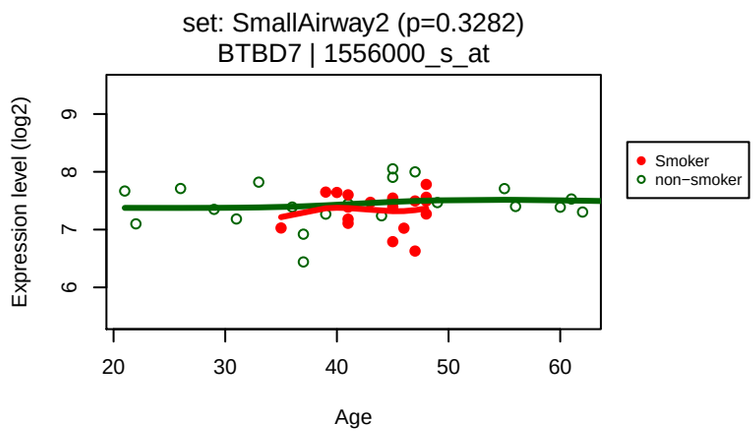
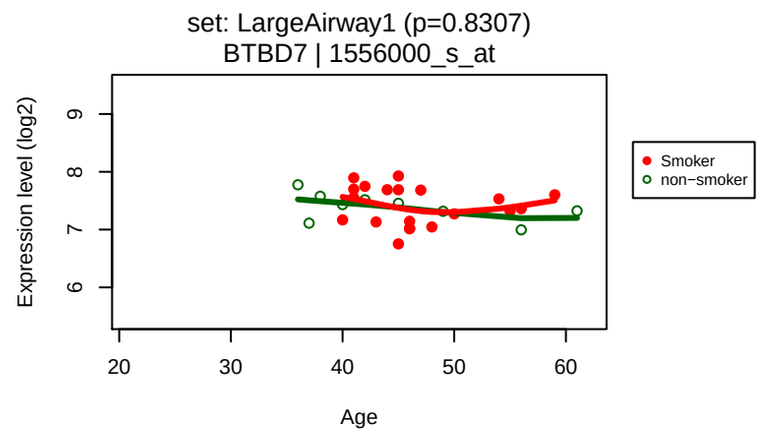

Gene expression of smokers and nonsmokers
gene: BTBD7 | 1556000_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1718
p=0.3282
p=0.5954
p=0.8307
p=0.5452
p=0.0161
p=0.4248
p=0.2719
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

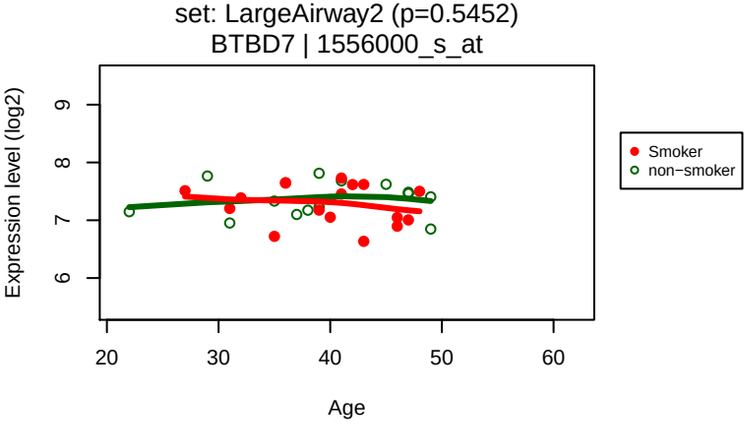

Gene expression ratio (smokers/nonsmokers)
gene: BTBD7 | 1556000_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1718
p=0.3282
p=0.5954
p=0.8307
p=0.5452
p=0.0161
p=0.4248
p=0.2719
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

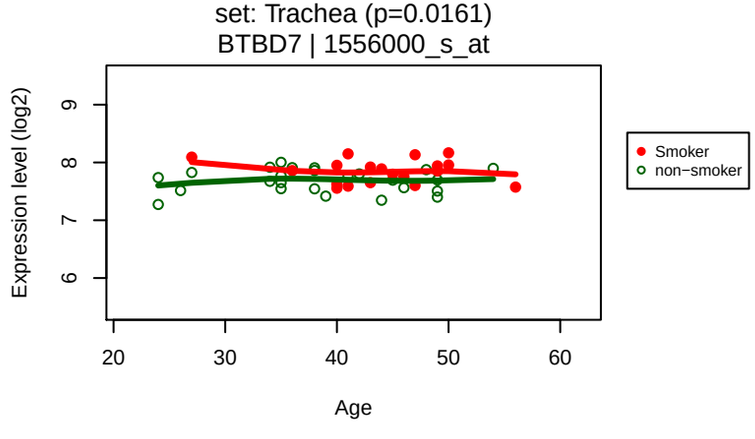

Gene expression ratio in Males
gene: BTBD7 | 1556000_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2725
p=0.6342
p=0.011
p=0.0704
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BTBD7 | 1556000_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.956
p=0.1881
p=0.5215
p=0.739
SmallAirway2
LargeAirway2
Trachea
Macrophage2

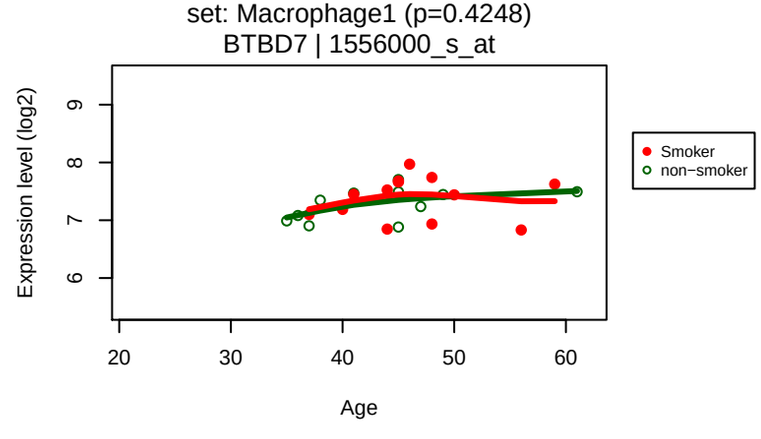

set: SmallAirway1 (p=0.1718)
BTBD7 | 1556000_s_at
Expression level (log2)
Age
Smoker
non−smoker

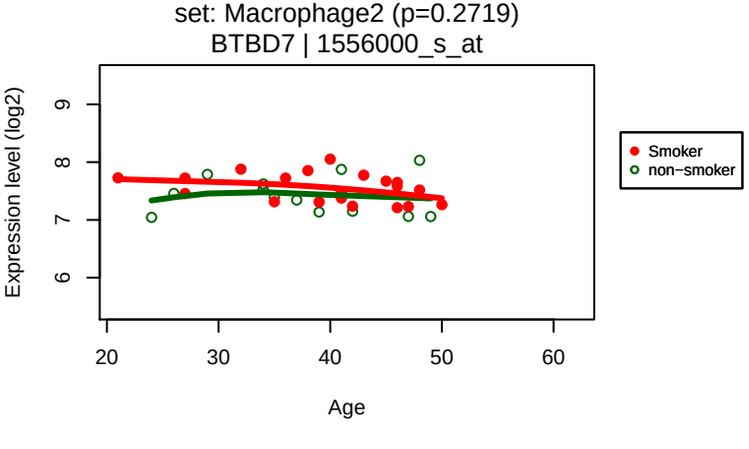

set: SmallAirway2 (p=0.3282)
BTBD7 | 1556000_s_at
Expression level (log2)
Age
Smoker
non−smoker

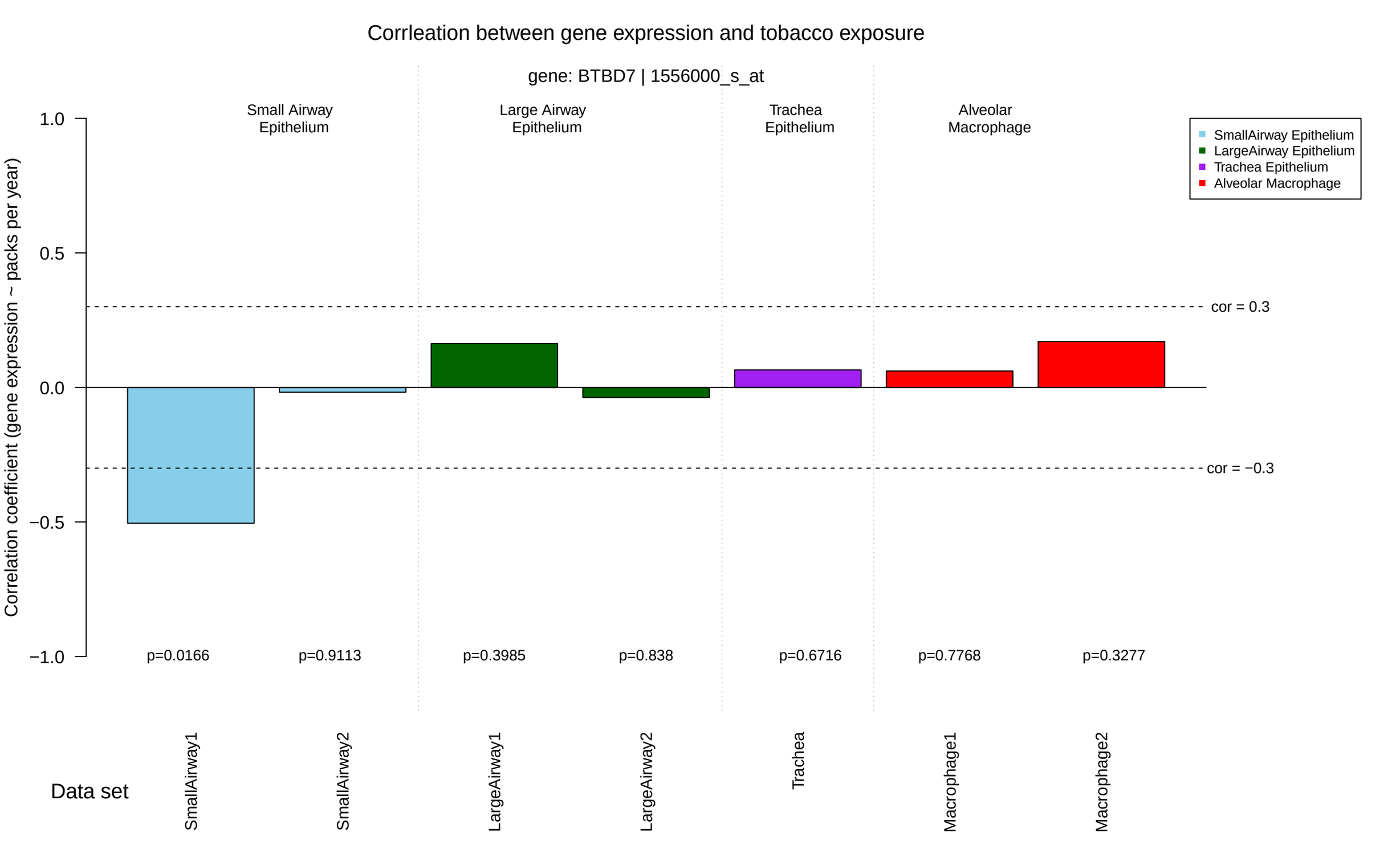

set: LargeAirway1 (p=0.8307)
BTBD7 | 1556000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5452)
BTBD7 | 1556000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0161)
BTBD7 | 1556000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4248)
BTBD7 | 1556000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2719)
BTBD7 | 1556000_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BTBD7 | 1556000_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0166
p=0.9113
p=0.3985
p=0.838
p=0.6716
p=0.7768
p=0.3277
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2